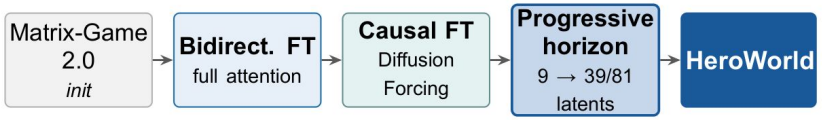
Long-Horizon Action-Conditioned Video Generation for Third-Person Interactive Environments

Presenter: Mingjia Huo (PhD@UCSD), Junda Su (PhD@UCSD), Will Lin (PhD@UCSD), Kaiqin Kong (MS@UCSD), Minshen Zhang (MS@UCSD), Shaoxiong Duan (BS@UCSD), Tajana Rosing (PI@UCSD), Hao Zhang (PI@UCSD)

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- **Problem & State of the Art**
  - Third-person action-conditioned video generation requires controlling a visible character through complex environments.
    - Over a 30-second rollout, a vanilla Self-Forcing baseline's appearance error (FVD-CLIP) grows by 58% (4.49→7.10).
- **Technical Approach**:
  - We develop a third-person, action-conditioned video world model trained on both simulator-generated and annotated YouTube videos, advancing controllable simulation.
  - We use a three-stage recipe (bidirectional finetuning, causal finetuning, and progressive-horizon self-forcing) for 30-second streaming rollouts.
  - We unify collected videos and simulator rollouts in a shared action space, enabling character-independent control.
- **Results**:
  - Benefiting from self-forcing with long rollouts, our method outperforms baselines [NeurIPS'25, ICLR'26], achieving the best visual fidelity (FVD-CLIP **3.2/4.2/4.5 over 30s**) and action-following performance.
  - Multi-domain training improves generalizability, boosting out-of-domain action-following by **5×**.
- **Showstopper**
  - Scaling to minute-long, high-resolution video generation requires a new hardware architecture to handle massive KV-cache storage and data movement.
- **Grand challenge application**: Deep Insights

a Three-stage training recipe



b Long-horizon rollouts (64 scenes, 0–30 s)

Method	FVD-CLIP ↓ 0–10 s	10–20 s	20–30 s	EPE ↓	VBench ↑ Quality
SF (vanilla)	4.49	6.87	7.10	5.54	0.791
LongLive (39)	3.30	4.26	4.78	4.70	0.801
Ours (9→39)	**3.20**	**4.16**	**4.48**	**4.55**	0.794
Ours (9→81)	3.95	4.76	5.08	5.23	**0.810**

c Out-of-domain control (88 unseen scenes)

Training data	OOD action cos ↑	VBench-Quality ↑
Single domain	0.145	0.773
+ Action scaling (×3)	0.450	0.830
4 real domains	0.250	0.791
+ Simulator rollouts	**0.772** (5.3×)	**0.852**

Publication: HeroWorld: Long-Horizon Action-Conditioned World Models for Third-Person Games. M. Huo, K. Kong, M. Zhang, J. Su, W. Lin, T. Rosing, H. Zhang, F2S@ICML 2026. (SRC # 476207)